

Assignment 2.

Question: A multiple-choice test consist of ten questions, each with four choices. Calculate the probability that a student who randomly guesses the answer to each question will get all of the question correct.
for each question having 4 choices $\rightarrow$ 1/4 answer correct

(Getting right choice): $\frac{1}{4}$ for each question

As there are 10 questions, the probability of getting all 10 correct are: $\left(\frac{1}{4}\right)^{10}$

Question: Determine the number of ways a jury of 6 men & 6 women can be chosen from an eligible pool of 12 men & 14 women

Given 6 men & women can be chosen from a eligible pool of 12 men & 14 women

combination formula: $nCr = \frac{n!}{(n-r)!r!}$ N: total people
r: chosen people

* If choose 6 men from 12.

$${}_{12}C_6 = \frac{12!}{6! \times 6!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{720} = \frac{665280}{720} = 924$$

$$\begin{array}{r} 16632 \\ 33264 \\ \hline 665280 \\ 720 \\ \hline 3618 \\ 39 \\ \hline 2773 \\ 3 \\ \hline 924 \end{array}$$

* If choose 6 women from 14:

$${}_{14}C_6 = \frac{14!}{8! \times 6!} = \frac{14 \times 13 \times 12 \times 11 \times 10 \times 9}{720} = \frac{3003}{1} = 3003$$

$$\begin{array}{r} 41 \\ 240 \\ 82 \\ \hline 3003 \end{array}$$

HW/VIVA - Import any 1 data set & do descriptive analysis.
with conclusion: case study

by using Multiplication Rule:-

$$3003 \times 924 = 2,774,772 \text{ ways}$$

Q:- In a high school graduating class of 100 students, 54 studies mathematics, 69 studied history & 35 studied both math & history. If one selected at random, find probability that

a) A student took maths & history

b) the student did not take either of these subjects

c) the student took history but not mathematics

Ans Given $n=100$, $|M|=54$, $|H|=69$, $|M \cap H|=35$

* Calculation of the student studying at least one subject

$$|M \cup H| = |H| + |M| - |M \cap H| = 69 + 54 - 35 = 88$$

$$(a) P(\text{student took maths } \cup \text{ history}) = \frac{88}{100} = 0.88$$

$$(b) P(\text{not taking any subject}) = \frac{n - |M \cup H|}{n} = \frac{100 - 88}{100} = \frac{12}{100} = 0.12$$

(c) $P(\text{take history not-maths})$

$$|H| = |H| - |M \cap H| = 69 - 35 = 34 \quad P(H \text{ only}) = \frac{34}{100} = 0.34$$

Q One bag contains 4 white ball & 3 black balls & a second bag contains 3 white balls & 5 black balls. One ball is drawn from the first bag & placed unseen in the second bag. What is probability that a ball now is drawn from second bag is black?

Given:- B_1 event for drawn a black ball from first bag
 B_2 event for drawn a black ball from IInd bag
 W_1 event for drawn a white ball from Ist bag
 W_2 event for drawn a white ball from IInd bag

$P(\text{Drawing white \& black ball from Bag 1})$

$$P(W_1) = \frac{4}{7} \quad P(B_1) = \frac{3}{7}$$

If white is transferred, the second bag now:

- 4 white balls (3 original + 1 transfer)
- 5 black balls.
- Probability of drawing black ball from the bag:

$$P(B_2|W_1) = \frac{5}{9}$$

If black ball is transferred

3 white & (5+1) black balls.

$P(\text{drawing a black ball from the second bag})$:

$$P(B_2|B_1) = \frac{6}{9}$$

by Law of Total Probability:-

$$P(B_2) = P(W_1)P(B_2|W_1) + P(B_1)P(B_2|B_1)$$

$$P(B_2) = \left(\frac{4}{7} \times \frac{5}{9} \right) + \left(\frac{3}{7} \times \frac{6}{9} \right)$$
$$\frac{20}{63} + \frac{2}{21}$$

$$\begin{array}{r|l} 3 & 13, 21 \\ \hline 7 & 21, 7 \\ \hline 3 & 3, 1 \\ \hline \end{array}$$

$$\frac{20 + 3 \times 2}{63} = \frac{20 + 6}{63} = \frac{26}{63}$$

Q:- The probability that a married man watches a certain TV show is 0.4, & probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7.

Find the probability that

- a married couple watches the show
- a wife watches the show given that her husband does;
- at least 1 member of married couple will watch the

$$P(\text{Married Men}) = 0.4$$

$$P(\text{Married Women}) = 0.5$$

$$P(\text{Married Men} / \text{Married Women}) = 0.7$$

a) $P(\text{both husband \& wife watch the show})$

$$P(M/W) = \frac{P(M \cap W)}{P(W)}$$

$$P(M \cap W) = P(M/W) \times P(W) = 0.7 \times 0.5 = 0.35$$

$$b) P(W|M) = \frac{P(W \cap M)}{P(M)} = \frac{0.35}{0.4} = 0.875$$

c) Probability (at least one member of couple watch show)

$$P(M \cup W) = P(M) + P(W) - P(M \cap W) = 0.4 + 0.5 - 0.35 = 0.55$$

Q A certain virus infects one in every 200 people. A test used for detect the virus in a person is positive 80% of the time if person has the virus & 5% of the time if the person does not have virus (5% called false positive). Let A be the event "the person is infected" & B be event "the person test positive".

a) Using Bayes Theorem, if a person test positive, determine the probability that the person is infected.

$$\text{Given: } P(A: \text{infected person}) = \frac{1}{200}$$

$$P(\text{not infected} : B) = 1 - \frac{1}{200} = \frac{199}{200}$$

$P(\text{positive test given the person is infected (True)})$

$$P(B|A) = 0.8$$

$P(\text{positive test given the person is not infected})$

$$P(B|A^c) = 0.05$$

a) Bayes Theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

Pl person is infected given they tested positive $P(A|B)$
for $P(B)$ by law of total probability

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$P(B) = (0.8 \times 0.005) + (0.005 \times 0.995)$$

$$= 0.004 + 0.004975 = 0.008975$$

$$P(A|B) = \frac{0.8 \times 0.005}{0.008975} = 0.044$$

b) Probability of person is NOT infected given they test
-ve $(A^c|B^c)$: $P(A^c|B^c) = \frac{P(B^c|A^c)P(A^c)}{P(B^c)}$

$$\text{As } P(B^c|A^c) = 1 - P(B|A^c) = 1 - 0.05 = 0.95$$

$P(B^c)$ is total probability of testing -ve:-

$$P(B^c) = P(B^c|A)P(A) + P(B^c|A^c)P(A^c)$$

$$= (0.2 \times 0.005) + (0.95 \times 0.995)$$

$$= 0.001 + 0.94525 = 0.94625$$

Now for ~~the~~ $P(A^c|B^c) =$

$$\frac{0.95 \times 0.995}{0.94625}$$

Q:- Calculate the total number of ways eight people
can be seated at a table of 8 seats

If they seated linearly: $8! = 40320$

If they seated in circular way $(8-1)! = 7! = 5040$

Q How many ways can you organise a class of 25 student of into groups of five?

Ans Divide them into 5 groups of 5 by using formula.

$$\frac{{}^{n}C_5 \times {}^{n-5}C_5 \times {}^{n-10}C_5 \times {}^{n-15}C_5 \times {}^{n-20}C_5}{5!}$$

$$n=25$$

$$5!$$

$${}^{25}C_5 = \frac{25!}{20!5!} = 53130$$

$${}^{15}C_5 = \frac{15!}{5!10!} = 3003$$

$${}^{20}C_5 = \frac{20!}{5!15!} = 15504$$

$${}^{10}C_5 = \frac{10!}{5!5!} = 252$$

$${}^5C_5 = 1$$

$$\frac{53130 \times 15504 \times 3003 \times 252 \times 1}{120} = 43589145600 \text{ (43 billion)}$$

Q If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary what is the probability that

(a) the dictionary is selected?

(b) 2 novels and 1 book of poems are selected?

Ans:-

$$5(\text{novels}) + 3(\text{poems}) + 1(\text{dictionary}) = 9 \text{ on shelf.}$$

So, Total number of ways to choose 3 books from is given by the combinations formula:

$${}^9C_3 = \frac{9!}{(9-3)!3!} = \frac{9!}{6!3!} = \frac{3 \times 2 \times 1}{6 \times 2} = 84$$

a) Dictionary Selected:

Ans: Total - dictionary = 8 books

$$\binom{8}{2} = \frac{8!}{(8-2)! 2!} = \frac{8!}{6! 2!} = \frac{8 \times 7}{2} = 28$$

$$P(\text{dictionary})_{\text{Selection}} = \frac{\binom{8}{2}}{\binom{9}{3}} = \frac{28}{84} = \frac{1}{3}$$

b) 2 novels & 1 book of poem:

$$\text{Novel: } 5C_2 = \frac{5!}{3! 2!} = \frac{5 \times 4}{2} = 10$$

$$\text{Poem: } 3C_1 = \frac{3!}{2! 1!} = \frac{3 \times 2}{2} = 3$$

by Multiplication method:- $10 \times 3 = 30$

$$P(2 \text{ novels and } 1 \text{ poem}) = \frac{\binom{5}{2} \times \binom{3}{1}}{\binom{9}{3}} = \frac{30}{84} = \frac{5}{14}$$

Q: If that we have fuse box containing 20 fuses, of which 5 are defective, if 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?

Total defective = 5

Non defective = $20 - 5 = 15$

* 2 selected random

* without replacing first

Probability that the first fuse selected is defective is

$$P(1st \text{ defective}) = \frac{5}{20} = \frac{1}{4}$$

Given: 1st fuse is defective, 19 fuses left. & of which $5 - 1 = 4$ is defective.

$P(2nd \text{ is defective} | 1st \text{ is defective})$

$$= \frac{4}{19} \left(\frac{\text{remaining fuse}}{\text{Total fuse}} \right)$$

$P(\text{both are defective}) = P(1st) \times P(2nd | 1st \text{ is defective})$

$$\frac{1}{4} \times \frac{4}{19} = \frac{1}{19}$$

Q In the senior year of a high school graduating class of 100 students, 42 studied mathematics, 68 studied psychology, 54 studied history, 22 studied both mathematics & history, 25 studied both mathematics & psychology, 7 studied history but neither math nor psychology, 10 studied all 3 sub. and 8 did not take any of the three. Randomly select a student from the class & find the probabilities of the following events.

- a) A person enrolled in psychology takes all 3 subjects.
b) A person not taking psychology is taking both history and mathematics. $U = 100$

$$|M: \text{Mathematics}| = 42$$

$$|P: \text{Psychology}| = 68$$

$$|H: \text{History}| = 54$$

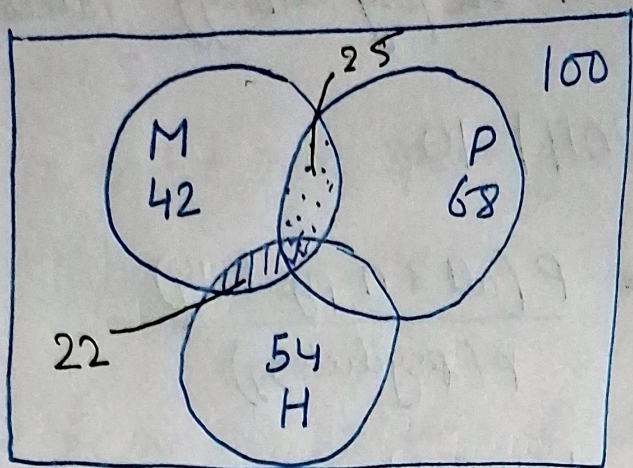
$$|M \cap H| = 22$$

$$|M \cap P| = 25$$

$$|H \cap M^c \cap P^c| = 7$$

$$|M \cap P \cap H| = 10$$

$$|M^c \cap P^c \cap H^c| = 8$$



Number of students in individual subject

$$M: |M| - (|M \cap H| - |M \cap P \cap H|) - (|M \cap P| - |M \cap P \cap H|) - (|M \cap P \cap H|) = 42 - 12 - 15 - 10 = 5$$

Let 'x' be the no. of student who studied history & psychology but not maths

$$P: |P| - (|M \cap P| - |M \cap P \cap H|) - (|H \cap P| - |M \cap P \cap H|) - |M \cap P \cap H| = 68 - 15 - x - 10 = 43 - x$$

H: 7 given

∴ equation

$$\text{Total} = \underset{\text{only}}{|M|} + \underset{\text{only}}{|P|} + \underset{\text{only}}{|H|} + |M \cap H| + |M \cap P| + |P \cap H| + |P \cap M \cap H| + |P \cap M \cap H|$$

$$\rightarrow 100 = 5 + (43 - x) + 7 + 12 + 15 + x + 10 + 8 = 100$$

To find x:

$$54 = 7 + 12 + x + 10$$

$$x = 54 - 29 = 25$$

∴ number of student studied history & psychology but not maths = 25

a) A person enrolled in psychology takes all 3 subjects.

$$|P| = 68, |M \cap N \cap H| = 10$$

$$P(\text{all} / \text{psychology}) = \frac{P(\text{all 3 in psychology})}{P(\text{psychology})}$$

$$= \frac{|M \cap N \cap H|}{|P|} = \frac{10}{68} = \frac{5}{34}$$

b) Student not taking psychology is taking math & history.

$$100 - 68 - 8 = 24$$

$$5 + 12 + 7 = 24$$

$$\frac{12}{24} = \frac{1}{2} = P(P \cap M^c \cap H^c)$$

$$= \frac{12}{32} = \frac{3}{8}$$

Q:- It is estimated that 50% of emails are spam emails. Some software has been applied filter these spam emails before they reach out your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%. Now if an email is detected as spam, then what is the probability that it is in fact a non-spam mail?

Ans: Let S be the event that an email is spam & N is non-spam email. Let D be the event that the software detects the email as spam.

$$P(S) = 0.50 (50\%)$$

$$P(N) = 1 - P(S) = 0.50$$

$$P(D|S) = 0.99$$

$$P(D|N) = 0.05$$

$P(\text{email is in fact non spam} \mid \text{it is detected spam})$

$$\text{i.e. } P(N|D) = \frac{P(D|N) P(N)}{P(D)} \quad \text{--- (1)}$$

Let $P(D)$ be the probability of spam
using total law of probability:

$$P(D) = P(D|S)P(S) + P(D|N)P(N)$$

$$= (0.99)(0.50) + (0.05)(0.05)$$

$$= 0.495 + 0.025$$

$$P(D) = 0.520 \quad \text{--- (11)}$$

put (2) in (1)

$$= \frac{(0.05)(0.50)}{0.520} = \frac{0.025}{0.520} = \frac{25}{520} = \frac{5}{104} \approx 0.048$$